



Pricing scheme design of ridesharing program in morning commute problem[☆]



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ABSTRACT

This paper examines the dynamic user equilibrium of the morning commute problem in the presence of ridesharing program. Commuters simultaneously choose departure time from home and commute mode among three roles: solo driver, ridesharing driver, and ridesharing rider. Considering the congestion evolution over time, we propose a time-varying compensation scheme to maintain a positive ridesharing ridership at user equilibrium. To match the demand and the supply of ridesharing service over time, the compensation scheme should be set according to the inconvenience cost functions and the out-of-pocket cost functions. When the price charged per time unit is higher than the inconvenience cost per time unit perceived by the ridesharing drivers, the ridesharing participants will travel at the center of peak hours and solo drivers will commute at the two tails. Within the feasible region with positive ridership, the ridesharing program can reduce the congestion and all the commuters will be better off. To support system optimum (SO), we derive a time-varying toll combined with a flat ridesharing price from eliminating queuing delay. Under SO toll, the ridesharing program can attract more participants and have an enlarged feasible region. This reveals that the commuters are more tolerant to the inconvenience caused by sharing a ride at SO because of the lower travel time. Compared with no-toll equilibrium, both overall congestion and individual travel cost are further reduced at SO.

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1. Introduction

With the development of the urban economy and the growth of population, severe traffic congestion has been observed in many cities, which results in heavy economic losses due to the increase in travel time and energy consumption. Various instruments have been proposed and implemented in order to alleviate congestion, such as congestion pricing and road space rationing. Recently, ridesharing has emerged as an alternative transport mode which can potentially reduce the number of vehicles on the road. The ridesharing platform, e.g., Uber and Lyft, matches drivers and riders in real time and coordinates the drivers to offer rides to travelers with similar itineraries. In return, the passengers will pay compensation to drivers according to pricing scheme provided by the matching agencies. Advanced technologies, such as smartphones and global position systems, enable the system to dynamically match shared rides on very short notice or even en-route

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in practice. Please refer to [Agatz et al. \(2012\)](#) for a comprehensive review in dynamic ridesharing and [Furuhata et al. \(2013\)](#) for an extensive discussion of future research directions.

The ridesharing literature is focused on optimizing the matching problem, i.e., determining the matches between drivers and riders in the system. The optimization models can be categorized according to the objectives. One of the conventional objectives is to minimize total travel distance or total travel time ([Agatz et al., 2010, 2011](#); [Wang, 2013](#)). Since the number of matched trips reflects the market penetration of the ridesharing program, some studies aim to maximize the number of the matched trips and participants ([Stiglic et al., 2015](#)). The goal of minimizing total waiting time is considered in [Guasch et al. \(2014\)](#). Both [Guasch et al. \(2014\)](#) and [Stiglic et al. \(2015\)](#) explore the effectiveness of using meeting points to facilitate matching. [Lee and Savelsbergh \(2015\)](#) optimize the matching of both ad hoc drivers and dedicated drivers with riders by associating each match with a weight according to its desirability. [Herbawi and Weber \(2012\)](#) study multiple objectives, e.g., maximizing the matched trips, and minimizing the total travel time. There are also other objectives including minimizing total passenger fares ([Santos and Xavier, 2013](#)) and the operating cost ([Lin et al., 2012](#)).

When the system allows a driver to pick up multiple riders, the simultaneous decisions in matching and routing become the necessary components in ridesharing platform. The similar problem exists in dial-a-ride problem for dispatching taxi service ([Dial, 2006](#)), which is essentially a pick-up and delivery problem with time window constraints. However, unlike the dial-a-ride problem in taxi booking, the drivers in ridesharing system may have their own destinations. [Wang et al. \(2016\)](#) extended the pick-up and delivery problem to optimize the routing considering the cost and time saving from High Occupancy Vehicle (HOV) lanes. Similarly, the idea of taxi-sharing is also explored in [Ma et al. \(2013, 2015\)](#) and [Hosni et al. \(2014\)](#). On the other hand, there are studies in facilitating riders to find optimal transfers between drivers, i.e., multi-hop in ridesharing, e.g., see [Herbawi and Weber \(2011a,b\)](#).

There is another stream of literature which examines the commuters' decision making when ridesharing is introduced to the existing road network and the aggregated impacts of ridesharing in traffic congestion ([Xu et al., 2014, 2015](#)). By assuming that driver can only pick up riders with same origin-destination (O-D) pair, [Xu et al. \(2014\)](#) extended the traffic assignment model with elastic demand to consider ridesharing as a new mode, and at user equilibrium, nobody can reduce his/her cost by changing route and mode choice. They found that, with a lower ridesharing price, it attracts more ridesharing participants. However, the equilibrium price does not monotonically increase or decrease the congestion level. [Xu et al. \(2015\)](#) relax the same O-D assumption in [Xu et al. \(2014\)](#) and formulate the problem with inelastic demand as a mixed complementarity problem. In their numerical experiments, they conclude that an increase in ridesharing price parameters will increase the number of solo drivers and therefore raise the level of congestion.

This study aims to examine the decision-making of the home-to-work commuters in the morning commute problem with ridesharing program. The analysis of travel patterns in morning commute problem is originated by [Vickrey \(1969\)](#). Given a desired arrival time at the workplace, the commuters choose the departure time from home and make the trade-off between the cost of travel time and the cost of schedule delay (when they cannot arrive at the workplace at the desired time). Accordingly, the temporal evolution of the congestion is explained by the commuters' departure time choices at dynamic user equilibrium at which nobody can reduce his/her own commute cost by unilaterally changing his departure time choice. Vickrey's bottleneck model has been extensively studied to examine the potential impacts of the travel demand management policies ([Small, 1982](#)). [Small \(2015\)](#) provides a comprehensive survey of the theory and the applications of bottleneck model in transportation economics. It has been extended to consider user heterogeneity ([Cohen, 1987](#); [Arnott et al., 1994](#); [Lindsey, 2004](#); [Small et al., 2005](#); [van den Berg and Verhoef, 2011](#); [Hall, 2013](#); [Liu et al., 2015](#); [Chen et al., 2015](#); [Wang and Du, 2016](#)), elastic demand ([Arnott et al., 1993](#)), multiple routes ([Arnott et al., 1990](#); [Adler and Cetin, 2001](#); [Liu and Nie, 2011](#)), and household activity ([Jia et al., 2016](#)). [Qian and Zhang \(2011\)](#) study the morning commute problem in the presence of HOV lanes when there are three modes available: transit, driving alone, and carpool. An extra constant cost is considered by carpool users, caused by the inconvenience at pick-up and drop-off. At static user equilibrium setting, [Yang and Huang \(1999\)](#) examine the design of congestion pricing combined with the operations of HOV lanes in a single O-D network. [Konishi and Mun \(2010\)](#) explore the welfare effects of introducing HOV lanes and converting HOV lanes to HOT lanes when the commuters have different carpool organization costs.

There is limited literature studying the ridesharing problem in the context of time-varying congestion. This paper assumes that, given a fixed total demand, the morning commuters will choose their departure time and mode among the three roles: solo driver, ridesharing driver, and ridesharing rider. Different from the carpool studies, we use a combined time-based and distance-based compensation scheme which is practiced in ridesharing industry. We find that a time-varying ridesharing pricing scheme should be set in order to maintain a positive ridesharing ridership at user equilibrium, depending on the costs of inconvenience and the out-of-pocket costs for ridesharing drivers and riders. Otherwise, the mismatch between the supply of ridesharing service and the demand of ridesharing service over time will occur, and nobody will choose ridesharing at the user equilibrium. In the presence of ridesharing program, the overall congestion is reduced given the proposed ridesharing pricing scheme. In addition, a time-varying system optimum (SO) toll and a flat ridesharing price are derived to support a stable equilibrium with minimum system cost. Combined with congestion pricing, the ridesharing program can attract more participants and have the larger feasible region with positive ridership because the queuing delay is eliminated and thereby the commuters are more tolerant to the inconvenience caused by sharing a ride. As we expected, compared with no-toll equilibrium (NTE), both overall congestion and individual travel cost are further reduced at SO. This study provides policy insights to support the regulation of ridesharing in the morning peak hours. A combined ridesharing pricing scheme and toll scheme is proposed for optimal system performance.

The remainder of the paper is organized as follows. Section 2 reviews the bottleneck model without ridesharing, and a new equilibrium is solved given the extra out-of-pocket cost defined. In Section 3, the equilibrium in the presence of ridesharing is derived. A time-varying pricing scheme is proposed in order to maintain a stable equilibrium with positive ridership. Section 4 proposes the SO toll with a constant ridesharing price. Section 5 demonstrates the numerical experiments with sensitivity analysis. Section 6 concludes our findings.

2. Review of bottleneck model without ridesharing

This section reviews the classic bottleneck model (Vickrey, 1969) without ridesharing. Please note that, different from the original bottleneck model, we add out-of-pocket cost here which results in a more general cost function. The out-of-pocket cost plays an important role in the decision making with ridesharing program (Section 3). To make this section consistent with the later parts, we first introduce and solve the bottleneck model without ridesharing by incorporating the out-of-pocket cost. In Vickrey's bottleneck model, there is a fixed number of N travelers who commute from home to work through a congested corridor during morning peak hours. It is assumed that a bottleneck is located at the exit of the corridor with limited capacity c . When the departure rate from home exceeds the capacity of the bottleneck, a queue will form and consequently commuters experience queuing delays on road. Therefore, the travel time along the corridor consists of two parts: (1) the fixed free-flow travel time T_0 , i.e., the travel time with no congestion, and (2) queuing delay. When commuters arrive at their work place, their schedule delay will be the difference between the actual and the desired arrival times. Each commuter chooses a departure time t (from home) so as to minimize his/her *generalized travel cost* $TC(t)$, which consists of the costs of travel delay and schedule delay. The notations in Section 2 are summarized in Table 1.

Following Vickrey's model, the assumptions in this study are listed here:

Assumption 1. The commuters have homogeneous preference. That is, they have identical value of time α , and identical unit cost of schedule delay (β for early arrival and γ for late arrival). And $\alpha > \beta$ for the existence of equilibrium (Lindsey, 2004).

Assumption 2. Let the desired arrival time be identical for all travelers, denoted by t^* .

Let $Q(t)$ denote the length of the queue that will be experienced by a commuter if she/he departs from home at time t . Then queuing time of the commuter is $\frac{Q(t)}{c}$. So her/his travel time is $TT(t) = T_0 + \frac{Q(t)}{c}$. And the schedule delay is $t^* - t - TT(t)$ for early arrival and $t + TT(t) - t^*$ for late arrival. The generalized travel cost includes the cost of travel time and the cost of schedule delay, which is denoted as $TC(t)$. Thus,

$$TC(t) = \begin{cases} \alpha TT(t) + \beta(t^* - t - TT(t)), & \text{if } t + T_0 + \frac{Q(t)}{c} \leq t^* \\ \alpha TT(t) + \gamma(t + TT(t) - t^*), & \text{if } t + T_0 + \frac{Q(t)}{c} > t^* \end{cases} \quad (1)$$

In order to make it comparable with the later section with ridesharing program, a more general travel cost function is defined here, which also includes out-of-pocket monetary costs, i.e., the fuel consumption, vehicle depreciation, insurance and so on (Litman, 2015). Note it is different from value of time (VOT). VOT is the opportunity cost of time, i.e., the amount that a traveler would be willing to pay to save a unit of time, which depends on traveler's preference. According to Litman (2015)¹, when road is congested, the out-of-pocket cost is related to travel time. We assume that the out-of-pocket cost is a function of both travel time and travel distance. The out-of-pocket cost $F(t)$ is defined as:

$$F(t) = c_t TT(t) + c_d d \quad (2)$$

where c_t denotes the monetary cost per unit time and c_d denotes the monetary cost per unit distance traveled. Hereafter, we will use $GC(t) = TC(t) + F(t)$ to denote the generalized travel cost. Compared with the original bottleneck model, we will first show that after taking consideration of out-of-pocket costs, the departure rate from home for arrival-early users at equilibrium will be decreased, and the departure rate for late-arrival users will be increased. Furthermore, the generalized travel cost is higher because of the additional out-of-pocket costs. However, the start time and the end time of congestion will not be affected.

Let t_q denote the start time of the bottleneck congestion and let $t_{q'}$ denote the end time of the bottleneck congestion respectively. After considering the out-of-pocket cost $F(t)$, the generalized travel cost of commuter who departs from home at time $t \in [t_q, t_{q'}]$ can be formulated as:

$$GC(t) = \begin{cases} \alpha(T_0 + \frac{Q(t)}{c}) + \beta(t^* - t - T_0 - \frac{Q(t)}{c}) + c_t(T_0 + \frac{Q(t)}{c}) + c_d v_f T_0, & \text{if } t + T_0 + \frac{Q(t)}{c} \leq t^* \\ \alpha(T_0 + \frac{Q(t)}{c}) + \gamma(t + T_0 + \frac{Q(t)}{c} - t^*) + c_t(T_0 + \frac{Q(t)}{c}) + c_d v_f T_0, & \text{if } t + T_0 + \frac{Q(t)}{c} > t^* \end{cases} \quad (3)$$

After simplifying Eq. (3), we have the following expressions:

¹ "Vehicle operating costs are about 40% higher on urban arterials than highways, and costs increase proportional to travel time when congestion reduces traffic speed to 30 mph on a highway or 20 mph on an arterial" (Litman, 2015).

Table 1
Notations without ridesharing.

α	Value of time
β	Unit cost of early schedule delay
γ	Unit cost of late schedule delay
c_t	Monetary cost per unit time in out-of-pocket cost function $F(t)$
c_d	Monetary cost per unit distance in out-of-pocket cost function $F(t)$
N	Total commuting demand, veh
T_0	Free flow travel time
c	Service rate or capacity of bottleneck, veh/h
t	Departure time from home
$Q(t)$	Queue length that will be experienced by a commuter if she/he departs from home at time t
$D(t)$	The cumulative number of departures from home up to time t
t^*	Desired arrival time
$\tilde{t}$	Departure time of the commuter who arrives at the workplace at exactly t^*
t_q	The start time of bottleneck congestion
t'_q	The end time of bottleneck congestion
$F(t)$	Out-of-pocket cost
$TC(t)$	Generalized travel cost in Vickrey's model
$GC(t)$	Generalized travel cost of commuter who departs from home at time t

$$GC(t) = \begin{cases} (\alpha - \beta + c_t + c_d v_f)T_0 + (\alpha - \beta + c_t) \frac{Q(t)}{c} + \beta(t^* - t), & \text{if } t + T_0 + \frac{Q(t)}{c} \leq t^* \\ (\alpha + \gamma + c_t + c_d v_f)T_0 + (\alpha + \gamma + c_t) \frac{Q(t)}{c} + \gamma(t - t^*), & \text{if } t + T_0 + \frac{Q(t)}{c} > t^* \end{cases} \quad (4)$$

When equilibrium is achieved, no one can reduce her/his generalized travel cost by changing her/his own departure time. Hence the first and last one arriving at the bottleneck have the same generalized travel cost:

$$\begin{cases} t_{q'} - t_q = \frac{N}{c} \\ \beta(t^* - t_q - T_0) = \gamma(T_0 + t_{q'} - t^*) \end{cases}$$

From the two equations above, we derive the formulas of t_q and $t_{q'}$:

$$\begin{cases} t_q = t^* - T_0 - \frac{\gamma}{\beta + \gamma} \frac{N}{c} \\ t_{q'} = t^* - T_0 + \frac{\beta}{\beta + \gamma} \frac{N}{c} \end{cases} \quad (5)$$

which is consistent with the start time and the end time of congestion in Vickrey (1969). Let $\tilde{t}$ denote the departure time of the commuter who arrives at the workplace at exactly t^* . We further impose the equilibrium condition across all the commuters. Specifically, the generalized travel cost is identical across departure time $t \in [t_q, t_{q'}]$:

$$\begin{aligned} \frac{dGC(t)}{dt} &= 0 \\ \Rightarrow \frac{dQ(t)}{dt} &= \begin{cases} \frac{\beta c}{\alpha - \beta + c_t}, & t \in [t_q, \tilde{t}) \\ \frac{-\gamma c}{\alpha + \gamma + c_t}, & t \in (\tilde{t}, t_{q'}] \end{cases} \end{aligned}$$

Given $Q(t_q) = 0$ and $Q(t_{q'}) = 0$, it has:

$$Q(t) = \begin{cases} \frac{\beta c}{\alpha - \beta + c_t} (t - t_q), & t \in [t_q, \tilde{t}] \\ \frac{\gamma c}{\alpha + \gamma + c_t} (t_{q'} - t), & t \in (\tilde{t}, t_{q'}] \end{cases} \quad (6)$$

Then we can derive the solution for $\tilde{t}$ by solving the following equations:

$$\begin{cases} Q(\tilde{t}) = \frac{\beta c}{\alpha - \beta + c_t} (\tilde{t} - t_q) \\ t^* = \tilde{t} + T_0 + \frac{Q(\tilde{t})}{c} \end{cases} \Rightarrow \tilde{t} = t^* - T_0 - \frac{\beta \gamma}{(\alpha + c_t)(\beta + \gamma)} \frac{N}{c}$$

Let $D(t)$ denote the cumulative number of departures from home up to time t . Then the departure rate from home, which is also equivalent to the arrival rate at the bottleneck, can be obtained

$$\dot{D}(t) = \dot{Q}(t) + c, \Rightarrow \dot{D}(t) = \begin{cases} \frac{\alpha + c_t}{\alpha - \beta + c_t} c, & t \in [t_q, \tilde{t}) \\ \frac{\alpha + c_t}{\alpha + \gamma + c_t} c, & t \in (\tilde{t}, t_{q'}] \end{cases} \quad (7)$$

From Eq. (7), we can easily verify that the departure rate in the proposed model is lower for the arrival-early commuters than the original bottleneck model, and is higher for the arrival-late commuters than the original bottleneck model. Combining Eqs. (3)–(6), the generalized travel cost $GC(t)$ can be obtained:

$$GC = (\alpha + c_t + c_d v_f) T_0 + \frac{\beta \gamma}{\beta + \gamma} \frac{N}{c} \quad (8)$$

Compared with the generalized travel cost in Vickrey's model, here it is increased by $(c_t + c_d v_f) T_0$, which is the out-of-pocket cost (Eq. (2)) associated with the free flow travel time T_0 and the constant travel distance.

The departure and arrival curves of equilibrium are plotted in Fig. 1.

3. Bottleneck model with ridesharing

In this Section, ridesharing program is introduced for the same model setting defined in Section 2. We will examine the user equilibrium with ridesharing participants when there are three options available for morning commuters, i.e., the solo driver, the ridesharing driver, and the ridesharing rider. We will first define the travel cost functions for each role respectively in Section 3.1. Given the cost function for each role, we will examine the existence of user equilibrium and thereby propose the pricing scheme so as to match ridesharing participants across time at user equilibrium in Section 3.2.

3.1. Model setting

When ridesharing program is available in the single bottleneck, a traveler makes decision on his preferable role (b) in the system. That is, the user chooses to drive alone (solo driver $b = 1$), serve as a ridesharing driver ($b = 2$), or a ridesharing passenger ($b = 3$). The definitions of the schedule delay cost, travel time cost and out-of-pocket cost remain the same as those introduced for bottleneck without ridesharing in Section 2. We will introduce several extra cost components which are associated with ridesharing program. Therefore, a more general form of travel cost is considered in this section.

Riding with a stranger decreases the comfort and the privacy for both drivers and riders. Therefore, ridesharing participants inevitably undergo different levels of inconvenience. To quantify the magnitude of inconvenience cost, we define it as a function of shared travel distance and travel time.

$$I_b(t) = \tau_b TT(t) + \delta_b d, \quad b = 2, 3 \quad (9)$$

where d denotes the length of shared distance, which is constant $d = v_f T_0$ in bottleneck model. And v_f denotes free flow speed. Since the travel time $TT(t)$ is a function of departure time from home t , the inconvenience cost function is also a function of t . Please note that, similarly, in Xu et al. (2015), the inconvenience cost is defined as a function of ridesharing driver flow and ridesharing rider flow on each link traversed. In the multi-modes study (transit, driving alone and carpool) by Qian and Zhang (2011), the inconvenience and the fuel costs are defined as constant. Zheng et al. (2014) empirically study the inconvenience cost for public transit commuters and estimate the inconvenience cost per time unit.

In ridesharing program, only drivers will bear the out-of-pocket cost and a rider pays a compensation to the matched driver. In fact, this is one of the major distinctions for ridesharing from traditional carpooling, in which carpoolers will share the out-of-pocket cost equally. Here we follow the commonly used pricing scheme in practice and define the price (the compensation) as a linear function of shared distance and travel time²:

$$p(t) = \omega_t TT(t) + \omega_d d \quad (10)$$

Given the inconvenience cost and the compensation defined above, the generalized cost $GC_b(t)$ for each role b ($b = 1, 2, 3$) can be obtained:

$$GC_1(t) = TC(t) + F(t) \quad (11)$$

$$GC_2(t) = TC(t) + I_2(t) + F(t) - p(t) \quad (12)$$

$$GC_3(t) = TC(t) + I_3(t) + p(t) \quad (13)$$

Here, $TC(t)$ and $F(t)$ are the same as defined in Section 2. Those ridesharing-related costs $I(t)$ and $F(t)$ are considered for both the drivers and the riders in the program. The notations in the presence of ridesharing are summarized in Table 2.

Besides Assumption 1 and 2, some mild assumptions are imposed here to further analyze the user equilibrium of morning commute problem with ridesharing program.

Assumption 3. The total demand is fixed while the demand for each role is determined by equilibrium. The same assumption is imposed by Xu et al. (2015).

² In the price scheme used by Uber, the passengers are charged according to the shared distance and travel time in addition to a base fare. If they cancel the proposed trip, a cancellation fee is required. <http://sanfrancisco.ubermovement.com/products/>.

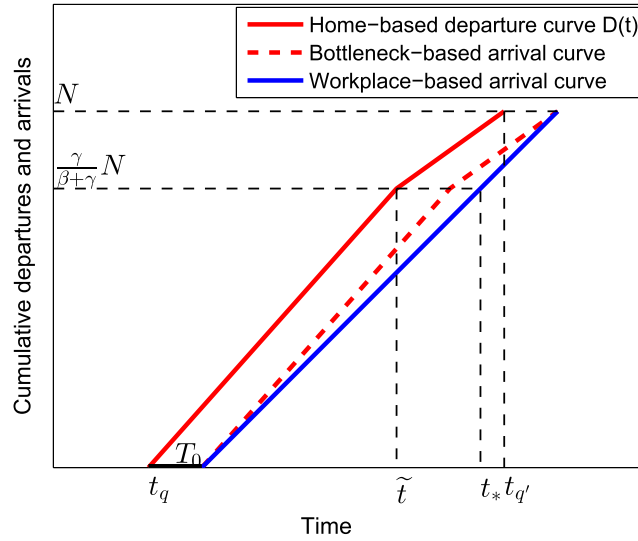


Fig. 1. User equilibrium without ridesharing.

Table 2

Notations with ridesharing.

b	Role, where $b = 1$ for solo drivers, $b = 2$ for ridesharing drivers and $b = 3$ for ridesharing riders
N_1	The demand of solo drivers
N_2	The demand of ridesharing drivers
N_3	The demand of ridesharing riders
d	Length of the link
t	Departure time from home
$TT(t)$	Travel time between the origin and destination
t_{q_1}	Departure time from home for the first solo driver
$t_{q'_1}$	Departure time from home for the last solo driver
t_{q_2}	Departure time from home for the first ridesharing driver (rider)
$t_{q'_2}$	Departure time from home for the last ridesharing driver (rider)
$TC(t)$	The sum of travel time cost and schedule delay cost
$p(t)$	The fare that a rider should pay for his trip
$I_2(t)$	Inconvenience cost for ridesharing drivers
$I_3(t)$	Inconvenience cost for ridesharing riders
$F_b(t)$	Out-of-pocket cost for role b and b takes values 1 and 2
$GC_b(t)$	Generalized travel cost for role b
ω_t, ω_d	Coefficient of travel time and distance for the fare
τ_b, δ_b	Coefficient of travel time and distance for inconvenience cost, $b = 2, 3$

Assumption 4. Each driver can only pick up one rider.

3.2. Equilibrium condition

Each commuter will choose her/his departure time and role in the system. At equilibrium, no one can reduce her/his individual travel cost by changing own decision. Substituting $TC(t)$, $F(t)$, $p(t)$, $I_2(t)$ and $I_3(t)$ into Eqs. (11)–(13), the generalized travel cost for each role can be derived. For solo drivers:

$$GC_1(t) = \begin{cases} (\alpha - \beta + c_t + c_d v_f)T_0 + (\alpha - \beta + c_t)\frac{Q(t)}{c} + \beta(t^* - t), & \text{if } t + T_0 + \frac{Q(t)}{c} \leq t^* \\ (\alpha + \gamma + c_t + c_d v_f)T_0 + (\alpha + \gamma + c_t)\frac{Q(t)}{c} + \gamma(t - t^*), & \text{if } t + T_0 + \frac{Q(t)}{c} > t^* \end{cases} \quad (14)$$

For simplicity, let $z_1 = c_t + c_d v_f$. Then, Eq. (14) becomes:

$$GC_1(t) = \begin{cases} (\alpha - \beta + z_1)T_0 + (\alpha - \beta + c_t)\frac{Q(t)}{c} + \beta(t^* - t), & \text{if } t + T_0 + \frac{Q(t)}{c} \leq t^* \\ (\alpha + \gamma + z_1)T_0 + (\alpha + \gamma + c_t)\frac{Q(t)}{c} + \gamma(t - t^*), & \text{if } t + T_0 + \frac{Q(t)}{c} > t^* \end{cases} \quad (15)$$

Further define $z_2 = \tau_2 - \omega_t + (\delta_2 - \omega_d)v_f$ and $z_3 = \tau_3 + \omega_t + (\delta_3 + \omega_d)v_f$. The generalized travel cost for ridesharing drivers and riders can be obtained, respectively:

$$GC_2(t) = \begin{cases} (\alpha - \beta + z_1 + z_2)T_0 + (\alpha - \beta + c_t + \tau_2 - \omega_t)\frac{Q(t)}{c} + \beta(t^* - t), & \text{if } t + T_0 + \frac{Q(t)}{c} \leq t^* \\ (\alpha + \gamma + z_1 + z_2)T_0 + (\alpha + \gamma + c_t + \tau_2 - \omega_t)\frac{Q(t)}{c} + \gamma(t - t^*), & \text{if } t + T_0 + \frac{Q(t)}{c} > t^* \end{cases} \quad (16)$$

$$GC_3(t) = \begin{cases} (\alpha - \beta + z_3)T_0 + (\alpha - \beta + \tau_3 + \omega_t)\frac{Q(t)}{c} + \beta(t^* - t), & \text{if } t + T_0 + \frac{Q(t)}{c} \leq t^* \\ (\alpha + \gamma + z_3)T_0 + (\alpha + \gamma + \tau_3 + \omega_t)\frac{Q(t)}{c} + \gamma(t - t^*), & \text{if } t + T_0 + \frac{Q(t)}{c} > t^* \end{cases} \quad (17)$$

As every commuter will simultaneously choose a departure time and a role in the system, the equilibrium is achieved when no one can reduce their individual cost by changing decisions. At equilibrium, all the travelers must have the identical commute cost. First, for each role b , the corresponding commute cost $GC_b(t)$ does not change across the departure time t . According to $\frac{dGC_b(t)}{dt} = 0$, the queuing rates for solo drivers remain constant:

$$\frac{dGC_1(t)}{dt} = 0 \Rightarrow \frac{dQ(t)}{dt} = \begin{cases} \frac{\beta c}{\alpha - \beta + c_t}, & \text{if } t + T_0 + \frac{Q(t)}{c} < t^* \\ \frac{-\gamma c}{\alpha + \gamma + c_t}, & \text{if } t + T_0 + \frac{Q(t)}{c} > t^* \end{cases} \quad (18)$$

Similarly, the queuing rates for ridesharing drivers and riders are obtained:

$$\frac{dGC_2(t)}{dt} = 0 \Rightarrow \frac{dQ(t)}{dt} = \begin{cases} \frac{\beta c}{\alpha - \beta + \tau_2 + c_t - \omega_t}, & \text{if } t + T_0 + \frac{Q(t)}{c} < t^* \\ \frac{-\gamma c}{\alpha + \gamma + \tau_2 + c_t - \omega_t}, & \text{if } t + T_0 + \frac{Q(t)}{c} > t^* \end{cases} \quad (19)$$

$$\frac{dGC_3(t)}{dt} = 0 \Rightarrow \frac{dQ(t)}{dt} = \begin{cases} \frac{\beta c}{\alpha - \beta + \tau_3 + \omega_t}, & \text{if } t + T_0 + \frac{Q(t)}{c} < t^* \\ \frac{-\gamma c}{\alpha + \gamma + \tau_3 + \omega_t}, & \text{if } t + T_0 + \frac{Q(t)}{c} > t^* \end{cases} \quad (20)$$

Please note that when the parameters τ_b , c_t , ω_t and ω_d are zero, the model will be reduced to the classic Vickrey's model without ridesharing, and queuing rates become $\beta c/(\alpha - \beta)$ for early arrival and $-\gamma c/(\alpha + \gamma)$ for late arrival.

We further examine the pricing scheme (ridesharing compensation) that ensures a positive matched ridesharing ridership at the user equilibrium in morning commute problem. It requires that (1) the ridesharing drivers and riders depart from home at the same rate across time (equivalently, the same $dQ(t)/dt, \forall t$), and (2) the drivers and the riders experience the same generalized travel costs. The proposed compensation scheme is presented in [Proposition 1](#).

Proposition 1 (Compensation scheme for the positive ridesharing ridership). *The ridesharing ridership is positive at user equilibrium, i.e. $N_2 = N_3 \geq 0$, if and only if*

- (1) $\omega_t = \frac{1}{2}[\tau_2 + c_t - \tau_3]$, which is the relationship of the time-based parameters. It ensures the ridesharing drivers and riders experience the same queuing rate.
- (2) $\omega_d = \frac{1}{2}[\delta_2 + c_d - \delta_3]$, which is the relationship of distance-based parameters. It ensures the equilibrium condition between the drivers and the riders, i.e., the same generalized travel costs of the two roles.

Proof.

1. Sufficiency. Because ridesharing driver and the corresponding rider share the same vehicle, they experience the same queuing rate. Thus, Eqs. (19) and (20) are equivalent for early arrival:

$$\frac{\beta c}{\alpha - \beta + \tau_2 + c_t - \omega_t} = \frac{\beta c}{\alpha - \beta + \tau_3 + \omega_t}$$

which gives the conclusion that

$$\omega_t = \frac{1}{2}(\tau_2 + c_t - \tau_3)$$

The same conclusion can be drawn by the equivalence between late arrivals for ridesharing drivers and riders.

Further, given the $GC_2(t)$ and $GC_3(t)$ for early or late arrivals, they should be equal to fulfill the requirement of long-run user equilibrium. Thus, we have $\omega_d = \frac{1}{2}(\delta_2 + c_d - \delta_3)$.

2. Necessity. Given $\omega_t = \frac{1}{2}(\tau_2 + c_t - \tau_3)$, we necessarily arrive at the same queuing rate for ridesharing drivers and riders. Additionally, if $\omega_d = \frac{1}{2}(\delta_2 + c_d - \delta_3)$ also holds, then ridesharing drivers and riders have the same travel cost. $\square$

Proposition 1 indicates that the pricing scheme of the ridesharing program should be set according to the inconvenience cost function and the out-of-pocket cost function in the morning commute problem. The time-based parameter ω_t is determined by the difference of the time-based parameters in inconvenience cost function $\tau_2 - \tau_3$, and the time-based parameter in the out-of-pocket cost function c_t . And the distance-based parameter ω_d is determined by the difference of the distance-based parameters in inconvenience cost function $\delta_2 - \delta_3$, and the distance-based parameter in the out-of-pocket cost function c_d . When condition (1) or (2) is not met, there will be no participant successfully matched during the peak hours even though the ridesharing program is available. As $\omega_t \neq \frac{1}{2}[\tau_2 + c_t - \tau_3]$, the two groups, i.e., the participating riders and the participating drivers, will choose to depart from home at two different rates in order to maintain equilibrium within each group. In other words, it implies a mismatch between the supply of ridesharing drivers and the demand from ridesharing riders. The imbalance would manifest as excess supply of service for some departure times, and excess demand for service at other times. Because of the assumed equilibrium conditions for decision making, all the unmatched ridesharing riders and drivers will choose to drive alone instead. When condition (1) is met, and $\omega_d \neq \frac{1}{2}[\delta_2 + c_d - \delta_3]$, the generalized travel costs are different between participating drivers and participating riders. Again, the commuters from the group with higher commute cost will switch to drive alone. It also results in excess of demand or supply, and thereby all the commuters will choose to drive alone at equilibrium.

By assuming compensation scheme is set as defined in **Proposition 1**, we proceed to examine the equilibrium solution, including the flows and the departure orders of the ridesharing participants and solo drivers. In the classic bottleneck analysis with heterogeneous preferences (α , β , and γ), [Arnott et al. \(1994\)](#) have shown that the departure orders are determined by the relative ratio $\beta/(\alpha - \beta)$ for early arrival and $-\gamma/(\alpha + \gamma)$ for late arrival. In this study, there are two groups, i.e., the solo drivers and the ridesharing commuters. **Proposition 1** has enforced the ridesharing drivers and riders to depart from home at the same rate, so that each driver can be matched with a rider to share a car. Therefore, by comparing the queuing rates of solo drivers and the ridesharing drivers/riders here, we can obtain the departure order of the two commuting groups: solo drivers and ridesharing participants.

Proposition 2 (The departure order at UE).

- Case 1: When $c_t > \tau_2 + \tau_3$, ridesharing participants depart from home at the center of peak hours.
- Case 2: When $c_t < \tau_2 + \tau_3$, solo drivers depart from home at the center of peak hours.
- Case 3: When $c_t = \tau_2 + \tau_3$, (3a) if $c_d = \delta_2 + \delta_3$, the departures of ridesharing participants and solo drivers are mixed, or (3b) if $c_d > \delta_2 + \delta_3$, all commuters choose ridesharing, or (3c) if $c_d < \delta_2 + \delta_3$, all commuters choose driving alone.

Proof. The departure order is determined by the $\dot{D}(t)$ or $\dot{Q}(t)$. The group with higher $\dot{D}(t)$ or $\dot{Q}(t)$ will depart at time slots closer to desired arrival time t^* . If $c_t > \tau_2 + \tau_3$ and $\omega_t = \frac{1}{2}(\tau_2 + c_t - \tau_3)$, we have $\frac{\beta c}{\alpha - \beta + c_t} < \frac{\beta c}{\alpha - \beta + \tau_2 + c_t - \omega_t} = \frac{\beta c}{\alpha - \beta + \tau_3 + \omega_t}$. Then the corresponding departure rate of solo drivers is lower than that of ridesharing participants, so ridesharing users will depart from home at the center of peak hours ([Arnott et al., 1994](#)).

Case 2 and 3 in **Proposition 2** can be obtained in a similar way. $\square$

Proposition 2 implies that the departure order depends on the relative magnitude of inconvenience cost for both a driver and a rider ($\tau_2 + \tau_3$) and out-of-pocket cost (c_t) per time unit. It reveals that, by choosing whether or not to participate in ridesharing program at departure time t , the commuters make trade-off between the inconvenience cost (the total inconvenience cost of two commuters in a vehicle) and out-of-pocket cost to minimize their individual generalized cost. Case 1 indicates that when the commuters perceive the inconvenience cost per time unit lower than the out-of-pocket cost per time unit, they will choose to participate at the center of the peak hours. Hence they would accept to travel at the center of peak hours and experience longer travel time. Otherwise, at Case 2 where the commuters perceive the inconvenience cost per time unit higher than the out-of-pocket cost per time unit, they will choose to participate at two tails of the peak hours where the queuing time is relatively lower. When $c_t = \tau_2 + \tau_3$, the inconvenience cost of sharing a ride per time unit will be exactly offset by the cost saving in the out-of-pocket cost. Thus, when condition $c_d = \delta_2 + \delta_3$ is also met, the commuters are indifferent to the two choices, i.e., to participate or not to participate. The two choices have the same isocost curve. Therefore, the departure time windows of the two groups are no longer two connected sets. Otherwise, when $c_d \neq \delta_2 + \delta_3$, it leads to corner solution in which either all commuters choose ridesharing or driving alone. In general, the departure order is not uniquely determined at Case 3.

We provide the equilibrium solution for the three cases below. That is, the equilibrium flow and the generalized cost for each role.

Case 1: $c_t > \tau_2 + \tau_3$

We first consider the situation $c_t > \tau_2 + \tau_3$ to derive the equilibrium solution in our model. When $c_t > \tau_2 + \tau_3$, the price charged per time unit is higher than the inconvenience cost per time unit perceived by ridesharing drivers ($\omega_t > \tau_2$) and ridesharing participants travel at the center of the peak hours and solo drivers travel at the two tails. As shown in [Fig. 2](#) (a), early-arrival solo drivers depart from home before ridesharing participants and late-arrival solo drivers depart from

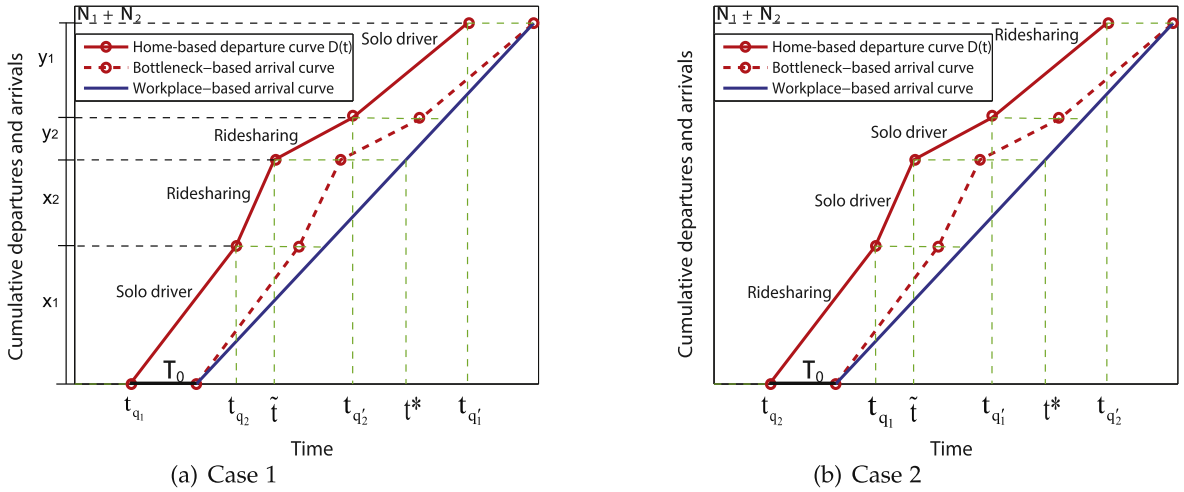


Fig. 2. User equilibrium with ridesharing.

home after ridesharing participants. Let t_{q_1} and $t_{q_1'}$ be the start and the end of the departure times from home for solo drivers, respectively. Let t_{q_2} and $t_{q_2'}$ be the start and the end of the departure times from home for ridesharing participants. The solid red line is the cumulative departure curve from home, and the solid blue line is the cumulative arrival curve at the workplace. Eqs. (18)–(20) have imposed the equilibrium conditions within each connected set, i.e., time interval $[t_{q_1}, t_{q_2}]$, $[t_{q_2}, \tilde{t}]$, $[\tilde{t}, t_{q_2'}]$, and $[t_{q_2'}, t_{q_1'}]$. In order to solve the equilibrium solution, extra conditions are required: (1) the generalized cost for each role is identical at equilibrium, and (2) for each role, the generalized cost for the two connected sets (the early-arrival set and the late-arrival set) is identical.

Given the queuing rate in Eqs. (18)–(20), the queue length ($Q(t)$) can be formulated as a continuous piece-wise function of t :

$$Q(t) = \begin{cases} \frac{\beta c}{\alpha - \beta + c_t} (t - t_{q_1}) & t \in [t_{q_1}, t_{q_2}] \\ \frac{\beta c}{\alpha - \beta + c_t + c_2 - \omega_t} (t - t_{q_2}) + Q(t_{q_2}) & t \in [t_{q_2}, \tilde{t}] \\ \frac{\gamma c}{\alpha + \gamma + c_t + c_2 - \omega_t} (t_{q_2'} - t) + Q(t_{q_2'}) & t \in [\tilde{t}, t_{q_2'}] \\ \frac{\gamma c}{\alpha + \gamma + c_t} (t_{q_1'} - t) & t \in [t_{q_2'}, t_{q_1'}] \end{cases} \quad (21)$$

Combining the generalized cost in Eqs. (14)–(17), and queue length in Eq. (21), the equilibrium conditions, i.e., the solo drivers have identical generalized cost at t_{q_1} and $t_{q_1'}$, give the relationship:

$$\begin{cases} CG_1(t_{q_1}) = GC_1(t_{q_1'}) \\ t_{q_1'} - t_{q_1} = \frac{N_1 + N_2}{c} \end{cases} \quad (22)$$

The start and the end of departure times from home for solo drivers (Fig. 2(a)) are available in Eq. (23).

$$\Rightarrow \begin{cases} t_{q_1} = t^* - T_0 - \frac{\gamma}{\beta + \gamma} \frac{N_1 + N_2}{c} \\ t_{q_1'} = t^* - T_0 + \frac{\beta}{\beta + \gamma} \frac{N_1 + N_2}{c} \end{cases} \quad (23)$$

Similarly, the equilibrium conditions that the ridesharing drivers and the solo drivers have identical generalized cost at t_{q_2} and $t_{q_2'}$, give:

$$\begin{cases} GC_1(t_{q_2}) = GC_2(t_{q_2}) \\ GC_1(t_{q_2'}) = GC_2(t_{q_2'}) \end{cases} \quad (24)$$

Furthermore, the equilibrium conditions that solo drivers have identical generalized cost at t_{q_1} , $t_{q_1'}$, t_{q_2} and $t_{q_2'}$, give:

$$\begin{cases} GC_1(t_{q_1}) = GC_1(t_{q_2}) \\ GC_1(t_{q_1'}) = GC_1(t_{q_2'}) \end{cases} \quad (25)$$

Combining Eqs. (23)–(25), the start and the end of departure times from home for ridesharing participants are:

$$\begin{cases} t_{q_2} = t^* - T_0 - \frac{\gamma}{\beta+\gamma} \frac{N_1+N_2}{c} + \frac{\alpha-\beta+c_t}{\beta} \frac{T_0[\tau_2-\omega_t+v_f(\delta_2-\omega_d)]}{\omega_t-\tau_2} \\ t_{q'_2} = t^* - T_0 + \frac{\beta}{\beta+\gamma} \frac{N_1+N_2}{c} - \frac{\alpha+\gamma+c_t}{\gamma} \frac{T_0[\tau_2-\omega_t+v_f(\delta_2-\omega_d)]}{\omega_t-\tau_2} \end{cases} \quad (26)$$

The commuter departing at time $\tilde{t}$ arrives at the workplace at time t^* , i.e.,

$$\tilde{t} + \frac{Q(\tilde{t})}{c} + T_0 = t^*$$

Together with Eq. (21), $\tilde{t}$ is obtained:

$$\tilde{t} = t^* - T_0 - \frac{\beta\gamma}{(\beta+\gamma)(\alpha+c_t+\tau_2-\omega_t)} \frac{N_1+N_2}{c} + \frac{T_0[\tau_2-\omega_t+v_f(\delta_2-\omega_d)]}{\alpha+c_t+\tau_2-\omega_t}$$

Let x_1 (y_1) be the number of early(late) arrivals of solo drivers. Let x_2 (y_2) denote the number of early(late) arrivals of ridesharing participants (Fig. 2(a)). The expressions of x_1, y_1, x_2 and y_2 are derived as follows:

$$\begin{aligned} x_1 &= c(t_{q_2} - t_{q_1}) \frac{\alpha+c_t}{\alpha-\beta+c_t} = c \frac{\alpha+c_t}{\beta} \frac{T_0(\tau_2-\omega_t+v_f(\delta_2-\omega_d))}{\omega_t-\tau_2} \\ y_1 &= c(t_{q'_1} - t_{q'_2}) \frac{\alpha+c_t}{\alpha+\gamma+c_t} = c \frac{\alpha+c_t}{\gamma} \frac{T_0(\tau_2-\omega_t+v_f(\delta_2-\omega_d))}{\omega_t-\tau_2} \\ x_2 &= c(\tilde{t} - t_{q_2}) \frac{\alpha+c_t+\tau_2-\omega_t}{\alpha-\beta+c_t+\tau_2-\omega_t} = \frac{\gamma}{\beta+\gamma} N_2 \\ y_2 &= c(t_{q'_2} - \tilde{t}) \frac{\alpha+c_t+\tau_2-\omega_t}{\alpha+\gamma+c_t+\tau_2-\omega_t} = \frac{\beta}{\beta+\gamma} N_2 \end{aligned}$$

Please note that the ratio of early arrivals to late arrivals for solo drivers and ridesharing participants are both:

$$\begin{aligned} \left[\frac{x_1}{y_1} \right]^{UE} &= \frac{\gamma}{\beta} \\ \left[\frac{x_2}{y_2} \right]^{UE} &= \frac{\gamma}{\beta} \end{aligned}$$

which remain the same as the original bottleneck model.

We can further derive the equilibrium solution by the following conditions:

$$N = N_1 + N_2 + N_3 \quad (27)$$

$$N_2 = N_3 \quad (28)$$

$$t_{q'_2} - t_{q_2} = \frac{N_1 + N_2}{c} - \frac{(\alpha+c_t)(\beta+\gamma)}{\beta\gamma} \frac{T_0[\tau_2-\omega_t+v_f(\delta_2-\omega_d)]}{\omega_t-\tau_2} \quad (29)$$

$$t_{q'_2} - t_{q_2} = \frac{N_2}{c} \quad (30)$$

Eq. (27) is the flow conservation equation and Eq. (28) ensures one-to-one pick up. The right-hand side of Eq. (29) is derived from Eq. (26). Eq. (30) is the capacity constraint. The equilibrium flow for each role is:

$$N_1 = \frac{c(\alpha+c_t)(\beta+\gamma)}{\beta\gamma} \frac{T_0[\tau_2-\omega_t+v_f(\delta_2-\omega_d)]}{\omega_t-\tau_2} \quad (31)$$

$$N_2 = N_3 = \frac{N}{2} - \frac{c(\alpha+c_t)(\beta+\gamma)}{2\beta\gamma} \frac{T_0[\tau_2-\omega_t+v_f(\delta_2-\omega_d)]}{\omega_t-\tau_2} \quad (32)$$

In addition, there exists an interior-solution region of N_1 , i.e., $N_1 \in (0, N)$, and an interior-solution region of N_2 and N_3 , i.e., $N_2 = N_3 \in (0, \frac{N}{2})$. The corner solutions are $N_1 = 0$ if $\tau_2 - \omega_t + v_f(\delta_2 - \omega_d) \leq 0$, or $N_1 = N$ if $\tau_2 - \omega_t + v_f(\delta_2 - \omega_d) \geq \frac{N(\omega_t - \tau_2)\beta\gamma}{c(\alpha+c_t)(\beta+\gamma)}$. The solution regions will be demonstrated by the numerical examples in Section 5. All the commuters at UE have the identical generalized travel cost:

$$GC = (\alpha+c_t+c_d v_f)T_0 + \frac{\beta\gamma}{\beta+\gamma} \frac{N_1+N_2}{c} \quad (33)$$

Since $N_1 + N_2 \leq N$, GC in the presence of ridesharing (Eq. (33)) is lower than that in the absence of ridesharing (Eq. (8)). So all the commuters are better off with the presence of ridesharing.

Case 2: $c_t < \tau_2 + \tau_3$

When $c_t < \tau_2 + \tau_3$, ridesharing participants depart from home at two tails and the solo drivers depart at the center of the peak hours (Fig. 2(b)). The critical times in this case are:

$$\begin{aligned}
t_{q_2} &= t^* - T_0 - \frac{\gamma}{\beta + \gamma} \frac{N_1 + N_2}{c} \\
t_{q'_2} &= t^* - T_0 + \frac{\beta}{\beta + \gamma} \frac{N_1 + N_2}{c} \\
t_{q_1} &= t_{q_2} + \frac{\alpha - \beta + c_t + \tau_2 - \omega_t}{\beta} \frac{z_2 T_0}{\omega_t - \tau_2} \\
t_{q'_1} &= t_{q'_2} - \frac{\alpha + \gamma + c_t + \tau_2 - \omega_t}{\gamma} \frac{z_2 T_0}{\omega_t - \tau_2}
\end{aligned}$$

Similarly, given conditions:

$$\begin{aligned}
N &= N_1 + N_2 + N_3 \\
N_2 &= N_3 \\
t_{q'_1} - t_{q_1} &= \frac{N_1 + N_2}{c} - \frac{(\alpha + c_t + \tau_2 - \omega_t)(\beta + \gamma)}{\beta\gamma} \frac{T_0[\tau_2 - \omega_t + v_f(\delta_2 - \omega_d)]}{\omega_t - \tau_2} \\
t_{q'_1} - t_{q_1} &= \frac{N_1}{c}
\end{aligned}$$

The UE solution is derived:

$$\begin{aligned}
N_1 &= N - \frac{2(\alpha + c_t + \tau_2 - \omega_t)(\beta + \gamma)}{\beta\gamma} \frac{cT_0[\tau_2 - \omega_t + (\delta_2 - \omega_d)v_f]}{\omega_t - \tau_2} \\
N_2 &= \frac{(\alpha + c_t + \tau_2 - \omega_t)(\beta + \gamma)}{\beta\gamma} \frac{cT_0[\tau_2 - \omega_t + (\delta_2 - \omega_d)v_f]}{\omega_t - \tau_2} \\
N_3 &= N_2
\end{aligned}$$

The generalized travel cost is

$$GC = (\alpha + c_t + \tau_2 - \omega_t + (c_d + \delta_2 - \omega_d)v_f)T_0 + \frac{\beta\gamma}{\beta + \gamma} \frac{N_1 + N_2}{c} \quad (34)$$

The derivation is omitted here. Again, the nonnegative constraints should be imposed, which gives us the interior-solution regions, i.e., $N_1 \in (0, N)$ and $N_2 = N_3 \in (0, \frac{N}{2})$, and the corner-solution regions, i.e., $N_1 = 0, N_2 = N_3 = N/2$ or $N_1 = N, N_2 = N_3 = 0$. With the condition $c_t < \tau_2 + \tau_3$ at Case 2 and the pricing scheme condition in Proposition 1, it gives $\tau_2 - \omega_t + (\delta_2 - \omega_d)v_f < 0$ when there are commuters participating in ridesharing, i.e., $N_2 > 0$. Therefore, we can verify that the generalized cost in Case 2 is also lower than that in the absence of ridesharing (Eq. (8)). Hence at Case 2, all the commuters are also better off after introducing ridesharing program.

Case 3: $c_t = \tau_2 + \tau_3$

When $\tau_2 + \tau_3 = c_t$, all the commuters are indifferent to the departure time. However, as explained below, the equilibrium solutions are generally not unique at case 3.

(a) When $c_d = \delta_2 + \delta_3$, the solo drivers and the ridesharing commuters should share the same iso-cost curve. According to the equilibrium definition, at any time t , $GC_1(t)$ (Eq. (15)) and $GC_2(t)$ (Eq. (16)) should be equivalent, which gives an extra condition for the existence of the interior solution for equilibrium ($N_1 > 0, N_2 > 0$), i.e., $c_d = \delta_2 + \delta_3$. For any $N_1 \in (0, N)$ and $N_2 = (N - N_1)/2$, it is an equilibrium solution. There are infinite number of equilibrium solutions. And the departure orders of solo drivers and ridesharing participants are mixed. In other words, the departure orders are not uniquely determined. The generalized travel cost has the same form as GC in Case 1 (Eq. (33)). When the interior solution for equilibrium exists ($N_1 > 0, N_2 > 0$), GC in Case 2 (Eq. (34)) is also reduced to Eq. (33).

(b) When $c_d > \delta_2 + \delta_3$, the ridesharing is a better choice than driving alone at any time t , so all commuters will choose ridesharing $N_1 = 0, N_2 = N/2$. The generalized travel cost is:

$$GC = \left(\alpha + c_t + c_d v_f - \frac{1}{2}(c_d - \delta_2 - \delta_3)v_f \right) T_0 + \frac{\beta\gamma}{\beta + \gamma} \frac{N}{2c}$$

(c) When $c_d < \delta_2 + \delta_3$, driving alone is a better choice than ridesharing at any time t , so all the commuters will choose to drive alone $N_1 = N, N_2 = 0$ and they experience a generalized travel cost:

$$GC = \left(\alpha + c_t + c_d v_f - \frac{1}{2}(c_d - \delta_2 - \delta_3)v_f \right) T_0 + \frac{\beta\gamma}{\beta + \gamma} \frac{N}{c}$$

4. System optimum

We solve the system optimum (SO) in this section. We will show that, at SO, the ridesharing participants should travel at the center of peak hours to reduce the system-wide schedule delay. A time-varying toll is derived to eliminate the queuing

delay. Under this tolling scheme, the ridesharing compensation scheme becomes a flat rate in the morning commute problem. And at SO, more commuters are matched given the proper support of compensation and SO toll schemes.

At the system optimum, queuing delay should be eliminated by toll and thereby the bottleneck-based arrival curve and the work-based arrival curve in Fig. 2(a) should overlap. In order to eliminate queuing delay, a time-varying toll ($Tol(t)$) should be implemented at the bottleneck. Please note that it is assumed that the toll is split equally between the rider and the driver here. Recall the definitions of generalized travel cost as Eqs. (14)–(17). When there is no queue, $Q(t) = 0$, the generalized travel costs are:

$$GC_1(t) = \begin{cases} (\alpha - \beta + z_1)T_0 + \beta(t^* - t) + Tol(t), & \text{if } t + T_0 \leq t^* \\ (\alpha + \gamma + z_1)T_0 + \gamma(t - t^*) + Tol(t), & \text{if } t + T_0 > t^* \end{cases} \quad (35)$$

$$GC_2(t) = \begin{cases} (\alpha - \beta + z_1 + z_2)T_0 + \beta(t^* - t) + \frac{Tol(t)}{2}, & \text{if } t + T_0 \leq t^* \\ (\alpha + \gamma + z_1 + z_2)T_0 + \gamma(t - t^*) + \frac{Tol(t)}{2}, & \text{if } t + T_0 > t^* \end{cases} \quad (36)$$

$$GC_3(t) = \begin{cases} (\alpha - \beta + z_3)T_0 + \beta(t^* - t) + \frac{Tol(t)}{2}, & \text{if } t + T_0 \leq t^* \\ (\alpha + \gamma + z_3)T_0 + \gamma(t - t^*) + \frac{Tol(t)}{2}, & \text{if } t + T_0 > t^* \end{cases} \quad (37)$$

We only study the case that ridesharing users travel at the center of peak hours and the solo drivers travel at the two tails. Considering the aggregate departures of ridesharing users, it is impossible for them to depart at the tails of peak hours at SO. As a ridesharing driver travels with a rider, the schedule delay cost will be further reduced when they choose to depart at times closer to t^* . Thus, the departure order is uniquely determined at SO. That is, ridesharing users travel at the center of peak hours. Furthermore, even if it is possible that the solo drivers travel at the center of peak hours, the SO can not be supported as a stable equilibrium by a time-dependent toll, because the toll will increase at a lower rate in the center of peak hours than that in the two tails. The derivation to rule out the case that solo drivers travel at the center is omitted here.

At SO, the departure rate from home is the same as bottleneck capacity. Thus, both the departure and arrival curves are straight lines with slope c . We follow the same approach in Liu and Nie (2011). By first assuming N_1 , N_2 , are given, the system cost minimization problem can be formulated as a minimization problem with decision variables x_1 , x_2 , y_1 and y_2 :

$$\min C \quad (38)$$

$$\text{s.t. : } x_1 + y_1 = N_1^s \quad (39)$$

$$x_2 + y_2 = N_2^s, \quad (40)$$

where the objective is the total cost, including total cost of schedule delay TC , total cost of inconvenience TI , and the total out-of-pocket cost TF . N_1^s denotes the number of solo drivers at SO and N_2^s denotes the ridership of ridesharing drivers at SO. Here the toll and the ridesharing compensation should not be accounted in C since the toll revenue and the compensation will circulate within transport system. These two constraints ensure that the number of drivers who can pass per hour is equivalent to the bottleneck capacity. x_1, x_2, y_1 and y_2 are defined as follows to facilitate the derivation.

$$x_1 = c(t_{q_2} - t_{q_1})$$

$$x_2 = c(t^* - T_0 - t_{q_2})$$

$$y_1 = c(t_{q_1} - t_{q_2})$$

$$y_2 = c(t_{q_2} + T_0 - t^*)$$

Therefore, all the cost components can be formulated as:

$$\begin{aligned} TC &= \alpha T_0(x_1 + y_1 + 2x_2 + 2y_2) + (\beta x_2^2 + \gamma y_2^2) + \frac{1}{2}\beta(x_1^2 + 2x_1x_2) \\ &\quad + \frac{1}{2}\gamma(y_1^2 + 2y_1y_2) \\ TF &= (c_t + c_d v_f)T_0(x_1 + y_1 + x_2 + y_2) \\ TI &= (\tau_2 + \tau_3 + (\delta_2 + \delta_3)v_f)T_0(x_2 + y_2) \\ \Rightarrow C &= TC + TF + TI = \bar{c} + (\beta x_2^2 + \gamma y_2^2) + \frac{1}{2}\beta(x_1^2 + 2x_1x_2) \\ &\quad + \frac{1}{2}\gamma(y_1^2 + 2y_1y_2) \end{aligned} \quad (41)$$

where $\bar{c} = \alpha T_0(x_1 + y_1 + 2x_2 + 2y_2) + TF + TI$. Obviously, given N_1^s and N_2^s , the out-of-pocket cost TF , and the inconvenience cost TI become constant when there is no queuing delay. And the first term in TC is also constant. Those constants can be replaced by $\bar{c}$ for simplicity. Given N_1^s and N_2^s , the first-order conditions for minimizing C with respect to x_1 and x_2 are:

$$\begin{aligned}\frac{\partial C}{\partial x_1} &= \beta x_1 + \beta x_2 - \gamma y_1 - \gamma y_2 = 0 \\ \Rightarrow x_1 + x_2 &= \frac{\gamma}{\beta + \gamma} (N_1^s + N_2^s)\end{aligned}\quad (42)$$

$$\begin{aligned}\frac{\partial C}{\partial x_2} &= 2\beta x_2 - 2\gamma y_2 + \beta x_1 - \gamma y_1 = 0 \\ \Rightarrow x_1 + 2x_2 &= \frac{\gamma}{\beta + \gamma} (N_1^s + 2N_2^s)\end{aligned}\quad (43)$$

Combining with Eqs. (39) and (40), we can obtain:

$$\Rightarrow \begin{cases} x_1 = \frac{\gamma}{\beta + \gamma} N_1^s \\ x_2 = \frac{\gamma}{\beta + \gamma} N_2^s \\ y_1 = \frac{\beta}{\beta + \gamma} N_1^s \\ y_2 = \frac{\beta}{\beta + \gamma} N_2^s \end{cases}$$

It is interesting to see that the ratio between the number of early arrivals and the number of late arrivals at SO is $x_1/y_1 = x_2/y_2 = \gamma/\beta$. With the presence of the ridesharing program, the ratio of early arrivals to late arrivals for each role at SO remains the same as the classic bottleneck results in the literature and the equilibrium without ridesharing in Section 2. This will also be demonstrated in Section 5.1.

Plugging x_1, x_2, y_1 and y_2 into C , the minimal system cost C^* can be formulated as function of N_1^s and N_2^s :

$$C^* = \frac{\beta\gamma}{2c(\beta + \gamma)} (N_2^s)^2 + (N_1^s + N_2^s)^2 + \alpha T_0 (N_1^s + 2N_2^s) + (c_t + c_d v_f) T_0 (N_1^s + N_2^s) + (\tau_2 + \tau_3 + (\delta_2 + \delta_3) v_f) T_0 N_2^s \quad (44)$$

The system optimal ridership can be solved by the following optimization program:

$$\begin{aligned} \min \quad & C^* \\ \text{s.t.} \quad & N_1^s + 2N_2^s = N, \\ & N_1^s \geq 0, \\ & N_2^s \geq 0, \end{aligned} \quad (45)$$

The Hessian matrix for program Eq. (45) is

$$H = \frac{\beta\gamma}{\beta + \gamma} \begin{bmatrix} \frac{1}{c} & \frac{1}{c} \\ \frac{1}{c} & \frac{2}{c} \end{bmatrix}$$

which is positive definite. So the SO problem has the unique solution. In order to solve this problem, we write the Lagrangian function of convex programming:

$$\begin{aligned} \min \quad & L = C^* + \lambda(N - N_1^s - 2N_2^s) \\ \text{s.t.} \quad & N_1^s \geq 0, \\ & N_2^s \geq 0, \end{aligned}$$

The KKT conditions of this problem are:

$$\begin{aligned} N_1^s \left(\frac{\beta\gamma}{2c(\beta + \gamma)} [2(N_1^s + N_2^s)] + (\alpha + c_t + c_d v_f) T_0 - \lambda \right) &= 0 \\ N_2^s \left(\frac{\beta\gamma}{2c(\beta + \gamma)} [2N_2^s + 2(N_1^s + N_2^s)] + (2\alpha + c_t + \tau_2 + \tau_3 + (c_d + \delta_2 + \delta_3) v_f) T_0 - 2\lambda \right) &= 0 \\ \lambda(N - N_1^s - 2N_2^s) &= 0 \end{aligned}$$

where $N_1^s \geq 0$ and $N_2^s \geq 0$.

Then riderships for solo driver and ridesharing driver at SO are:

$$N_1^s = \frac{2(\beta + \gamma)}{\beta\gamma} c T_0 [\tau_2 - \omega_t + (\delta_2 - \omega_d) v_f] \quad (46)$$

$$N_2^s = \frac{N}{2} - \frac{\beta + \gamma}{\beta\gamma} c T_0 [\tau_2 - \omega_t + (\delta_2 - \omega_d) v_f] \quad (47)$$

The nonnegativity of N_1^s and N_2^s requires $(\tau_2 - \omega_t + (\delta_2 - \omega_d) v_f) \geq 0$. Recall the equilibrium solution of Case 1 in Eq. (31) and pricing scheme $\omega_t = \frac{1}{2}[\tau_2 + c_t - \tau_3]$ in Proposition 1:

$$N_1^s = \frac{2(\omega_t - \tau_2)}{\alpha + c_t} N_1 = \frac{c_t - \tau_2 - \tau_3}{\alpha + c_t} N_1 < N_1$$

The implications here are, first, when $N_1 \in (0, N)$ at NTE Case 1 $c_t > \tau_2 + \tau_3$, the number of solo drivers would be reduced at SO. Furthermore, with larger time-based parameters in inconvenience functions (τ_2 and τ_3), the relative ratio of solo driver flows between SO and NTE, $\frac{c_t - \tau_2 - \tau_3}{\alpha + c_t}$, will be smaller. And the feasible region with positive ridership in ridesharing will be enlarged at SO. Those will be illustrated in the sensitivity analysis at Section 5.3.

When $N_1 \in (0, N)$ at NTE Case 2, similar to Case 1, at the interior-solutions region, it has $\tau_2 - \omega_t + (\delta_2 - \omega_d)v_f < 0$. It results in a corner solution at SO. That is, $N_1^s = 0$ and the number of solo drivers is eliminated at SO in this case. Therefore, more commuters will participate in ridesharing at SO. Different from Case 1, all the commuters will participate in ridesharing at SO if it falls into the interior-solution region of Case 2 at NTE. Therefore, the interior-solutions region at NTE corresponds to corner-solutions region at SO. And, $N_1^s/N_1 = 0$ indicates that the relative ratio of solo drivers between SO and NTE at Case 2 remains zero, which is lower than that at Case 1. It is interesting to see that if ridesharing participants traveling at the tails of peak hours at NTE (i.e., Case 2), the participation rate at SO ($N_2^s = N_3^s = N/2$) is irrelevant to the inconvenience cost associated with ridesharing.

In order to get the formulation of toll scheme $Tol(t)$, we need to derive the marginal cost (mc_1 or mc_2) and let generalized travel cost (Eqs. (35)–(37)) be equal to marginal cost. As mentioned, the ridesharing drivers and riders will share the $Tol(t)$ evenly if they travel in the same vehicle. The marginal cost of solo drivers and ridesharing drivers are obtained by the first-order derivative of C^* .

$$\begin{aligned} \frac{\partial C^*}{\partial N_1^s} &= \frac{\beta\gamma}{\beta + \gamma} \frac{N_1^s + N_2^s}{c} + (\alpha + c_t + c_d v_f) T_0 = mc_1 \\ \frac{\partial C^*}{\partial N_2^s} &= \frac{\beta\gamma}{\beta + \gamma} \frac{N_1^s + 2N_2^s}{c} + (2\alpha + c_t + c_d v_f + \tau_2 + \tau_3 + (\delta_2 + \delta_3)v_f) T_0 = mc_2 \end{aligned}$$

The time-varying toll scheme is derived as followings. And the toll scheme $Tol(t)$ is also shown in Fig. 3.

$$Tol(t) = \begin{cases} \beta(t - t_{q_1}) & t \in [t_{q_1}, t_{q_2}] \\ 2\beta(t - t_{q_2}) + \frac{\beta\gamma}{\beta + \gamma} \frac{N_1^s}{c} & t \in (t_{q_2}, \tilde{t}] \\ 2\gamma(t_{q'_2} - t) + \frac{\beta\gamma}{\beta + \gamma} \frac{N_1^s}{c} & t \in (\tilde{t}, t'_{q_2}] \\ \gamma(t'_{q_1} - t) & t \in (t'_{q_2}, t'_{q_1}] \end{cases} \quad (48)$$

When the system optimum is achieved, the values of t_{q_1} , $t_{q'_1}$, t_{q_2} and $t_{q'_2}$ are different from UE with ridesharing in Section 3. The reason is that the role assignment has changed when introducing $Tol(t)$ to get the system optimum. Specifically, tolling leads to an increase in the number of ridesharing participants (as $N_1^s < N_1$ shown above), and hence a reduction in the duration of the overall travel period. In this situation,

$$t_{q_1} = t^* - T_0 - \frac{\gamma}{\beta + \gamma} \frac{N_1^s + N_2^s}{c} \quad (49)$$

$$t_{q_2} = t^* - T_0 - \frac{\gamma}{\beta + \gamma} \frac{N_2^s}{c} \quad (50)$$

$$t_{q'_2} = t^* - T_0 + \frac{\beta}{\beta + \gamma} \frac{N_2^s}{c} \quad (51)$$

$$t_{q'_1} = t^* - T_0 + \frac{\beta}{\beta + \gamma} \frac{N_1^s + N_2^s}{c} \quad (52)$$

Eq. (48) has similar patterns as toll in Amott et al. (1990) and Liu and Nie (2011). But the critical times are different because of cost considerations. Now the generalized travel cost are

$$GC_1^{SO} = (\alpha + c_t + c_d v_f) T_0 + \frac{\beta\gamma}{\beta + \gamma} \frac{N_1^s + N_2^s}{c} \quad (53)$$

$$GC_2^{SO} = (\alpha + c_t + \tau_2 - \omega_t + (c_d + \delta_2 - \omega_d)v_f) T_0 + \frac{\beta\gamma}{\beta + \gamma} \frac{0.5N_1^s + N_2^s}{c} \quad (54)$$

$$GC_3^{SO} = (\alpha + \tau_3 + \omega_t + (\delta_3 + \omega_d)v_f) T_0 + \frac{\beta\gamma}{\beta + \gamma} \frac{0.5N_1^s + N_2^s}{c} \quad (55)$$

Given Eqs. (46) and (47), $N_1^s + N_2^s$ is lower than $N_1 + N_2$ at Case 1, which is calculated from Eqs. (31) and (32). Considering both the interior-solution region and the corner-solution region, the total traffic is reduced or remains the same. Thus, the generalized travel cost at SO (Eq. (53)) is lower than the generalized travel cost at NTE, i.e., Eq. (33). The commuters are

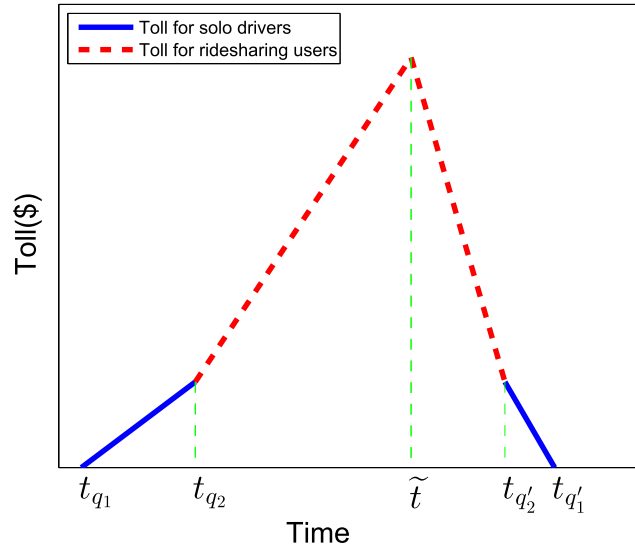


Fig. 3. Time-varying toll that supports system optimum.

equally well off when SO has corner solutions. Similarly, we can verify that the solo driver flow and the generalized cost at SO is reduced or at least remains the same compared with NTE at Case 2 and Case 3. Therefore, we conclude that the commuters at SO are further better off or at least equally well off at SO with ridesharing.

After implementing tolling scheme, generalized travel cost should be identical across solo drivers, ridesharing drivers and ridesharing riders at equilibrium. Therefore, given $GC_2^{SO}(t) = GC_3^{SO}(t)$ (Eqs. (54) and (55)), we have the following proposition:

Proposition 3 (Compensation scheme to support SO with positive ridesharing ridership). *To support the SO ridesharing ridership as equilibrium, the cost parameters should satisfy $2\omega_t + 2\omega_d v_f = \tau_2 + c_t - \tau_3 + (\delta_2 + c_d - \delta_3) v_f$.*

Obviously, the condition at SO above is a relaxation of the two conditions in Proposition 1. Rather than requiring ω_t and ω_d as derived in Proposition 1, the setting of ridesharing pricing parameters is more flexible. Any combination of ω_t and ω_d that meet condition in Proposition 3 can support SO. Because there is no queuing delay at SO, each commuter experiences a same travel time T_0 . According to the definition of ridesharing compensation p and Proposition 3, p equals to $(\omega_t + \omega_d v_f) T_0$, which is a constant. Namely, all the riders at SO will pay a flat price regardless of departure time choices, which will be demonstrated in the numerical section later.

Following from Eqs. (46) and (47), the boundary conditions for the existence of the interior solutions, i.e., $N_1 \in (0, N)$ or $N_2 \in (0, \frac{N}{2})$, are derived: $0 \leq \tau_2 + \tau_3 - c_t + (\delta_2 + \delta_3 - c_d) v_f < \frac{N}{cT_0} \frac{\beta\gamma}{\beta+\gamma}$. Based on those conditions, we draw some conclusions by conducting sensitivity analysis in Section 5.3.

5. Numerical experiments

Numerical experiments are conducted to demonstrate our analytical results. The sensitivity analysis, i.e., the results by varying parameter inputs, will also be reported.

5.1. No-toll equilibrium and system optimum

We adopt $\alpha = \$5$, $\beta = \$3.05$, $\gamma = \$11$ from Arnott et al. (1993). The desired arrival time for all the travelers is $t^* = 8.5$. When it is not congested, a commuter can reach the destination within free-flow travel time $T_0 = 0.5$ hour at the free-flow speed $v_f = 40$ mile/hour. The capacity of the bottleneck is $c = 600$ veh/hour and the total demand is $N = 2000$. All the input parameters are listed in Table 3. Without ridesharing program, the UE solutions are shown in Fig. 4(a).

The input parameters are set by referring to the related empirical studies. Zheng et al. (2014) study the transit attractiveness and estimate the value of convenience associated with unit of travel time (τ_2, τ_3) is from 0.48 to 5.6 \$/hour.³ American Automobile Association (AAA) estimates the out-of-pocket cost per mile is about \$0.37 in 2016⁴, i.e., c_d here. It is estimated that fuel consumption increases 40% in heavy congested conditions (Litman, 2015). Thus the out-of-pocket cost per hour is set to be

³ Note that the values of τ_2 and τ_3 are unlikely to be indicative of preference for ridesharing, due to the difference between travelers' perception of inconvenience towards ridesharing and public transit.

⁴ AAA, Your Driving Costs, 2016 Edition. Retrieved from <http://exchange.aaa.com/wp-content/uploads/2016/04/2016-YDC-Brochure.pdf>.

Table 3
Input for single bottleneck.

Parameter	Value	Unit
α	5	\$/h
β	3.05	\$/h
γ	11	\$/h
τ_2	2	\$/h
τ_3	0.3	\$/h
δ_2	0.25	\$/mile
δ_3	0.15	\$/mile
c_t	2.96	\$/h
c_d	0.37	\$/mile
v_f	40	mile/h
c	600	veh/h
N	2000	veh
T_0	0.5	h
t^*	8.5	am

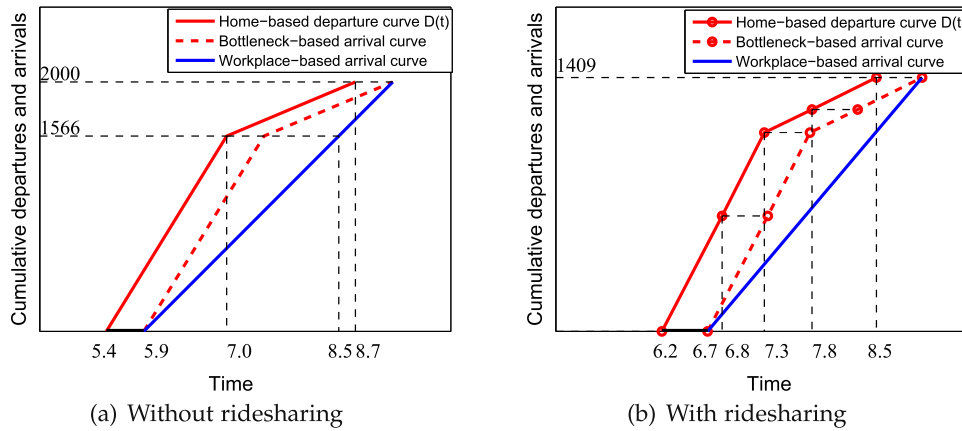


Fig. 4. Equilibrium with and without ridesharing program.

$c_t = \$2.96$. In our numerical experiments, c_t and c_d are considered fixed, i.e., assuming there is no fluctuation in market gasoline price.

Given the input in Table 3, the ridesharing compensation p can be determined by Proposition 1. The unit price of travel time is $\omega_t = \$2.33/\text{h}$ and the unit price of shared distance is $\omega_d = \$0.235/\text{mile}$. Please note the values taken are lower than the rates used by UberX, because ridesharing is not operated for profit in this study.

Recall Proposition 2, ridesharing users travel at the center of peak hours when $c_t > \tau_2 + \tau_3$. The user equilibrium with ridesharing is shown in Fig. 4(b), with 1182 commuters participating in ridesharing. The overall congestion time window is $[6.2, 8.5]$. This time duration equals to $\frac{N_1 + N_2}{c} = 2.3\text{h}$. When ridesharing is not available, the congestion time window is $[5.4, 8.7]$ (Fig. 4(a)). By comparison, it shows that ridesharing helps to shorten congestion duration given the fixed total demand. The congestion time windows for ridesharing participants are $[6.8, 7.3]$ for early arrivals and $[7.3, 7.8]$ for late arrivals (the numbers are rounded). For solo drivers, their departure time windows are $[6.2, 6.8]$ for early arrivals and $[7.8, 8.5]$ for late arrivals. The number of early-arrival solo drivers is 641 and the number of late-arrival solo drivers is 178. While for ridesharing participants, there are 463 ridesharing drivers arriving early and 128 ridesharing drivers arriving late. Hence, as we show in Section 3, the ratio of early arrivals to late arrivals is equivalent to $\gamma/\beta = 3.61$ for both solo drivers and ridesharing participants.

At SO, congestion pricing is introduced to eliminate queuing delay. Here a time-varying toll is derived and SO solutions are solved. The SO are shown in Fig. 5.

The number of ridesharing participants at SO in Fig. 5 is 1932. As discussed analytically at Section 4, given time-varying SO toll (Eq. (48)), more commuters will join ridesharing program compared with no-toll equilibrium in Fig. 4(b). Additionally, the peak time window further shrinks under the congestion pricing policy. Originally, the departure time interval is $[6.2, 8.5]$ at no-toll equilibrium, while it is $[6.65, 8.37]$ at SO. Specifically, the arrival-early departure time window at SO is $[6.65, 6.74]$ for solo drivers and $[6.74, 8]$ for ridesharing users. The arrival-late departure time window at SO is $[8.35, 8.37]$ for solo drivers and $[8, 8.35]$ for ridesharing drivers. The ratios of early arrivals to late arrivals are $\gamma/\beta = 3.61$ for both solo drivers and ridesharing participants.

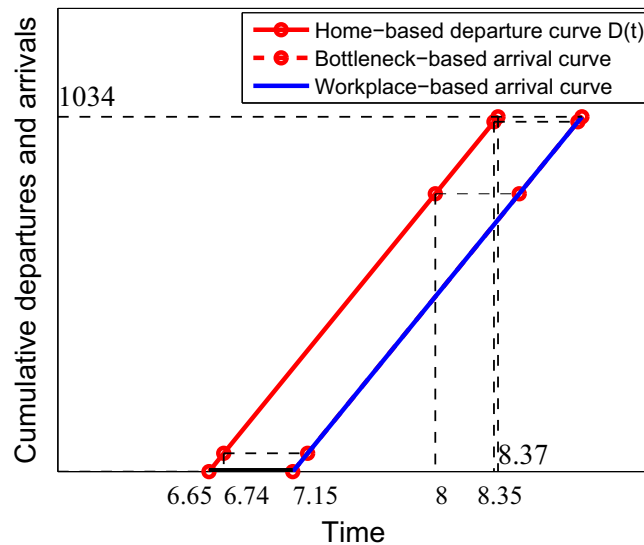


Fig. 5. SO solutions with ridesharing.

The system cost, defined in Eq. (41), is calculated for the three scenarios: (a) the equilibrium without ridesharing, (b) the equilibrium with ridesharing, and (c) the system optimum. The system cost measures the system performance, including the cost of queuing delay, the cost of schedule delay and other costs (the out-of-pocket cost and inconvenience cost). The results are available in Table 4. Please note the free flow travel time is not counted in this table. As shown in Table 4, compared with equilibrium without ridesharing, the ridesharing program can significantly reduce both the queuing delay (40.9%) and the schedule delay (49.6%). And at SO, the queuing delay is completely removed and system cost is improved by 37.8% from the equilibrium without ridesharing. Please note that, different from the classic bottleneck model, the cost of queuing delay is not equivalent to the cost of schedule delay at (a) the equilibrium without ridesharing. It is because the out-of-pocket cost is introduced in the generalized cost function at Section 2. In our model, the total queuing delay is lower than the total schedule delay at Scenario (a).

5.2. Ridesharing pricing schemes

The ridesharing price p is defined as a function of travel time TT and shared distance d as in Section 3.1. The distance is a constant which is a product of free-flow travel time and free-flow speed, i.e., $d = v_f T_0$. Furthermore, travel time is defined as $TT(t) = T_0 + \frac{Q(t)}{c}$, thus the ridesharing price is actually $p(t) = (\omega_t + \omega_d v_f) T_0 + \omega_t \frac{Q(t)}{c}$. Therefore, $p(t)$ is a piece-wise linear function of t . Recall the pricing scheme is determined by conditions in Proposition 1, so the time parameter is $\omega_t = \$2.33/h$ and the distance parameter is $\omega_d = \$0.235/mile$. The time-varying price $p(t)$ is shown in Fig. 6(a).

As revealed in Section 3, to successfully match ridership in ridesharing program, the price at UE should be set proportional to the congestion. $p(t)$ reaches the highest value when the congestion is the most severe, which happens at departure time from home $\tilde{t}$. Fig. 6(a) shows the range of price is $[5.8, 7.5]$. Without setting the pricing scheme properly, it is difficult to attract the ridesharing participants across time in the morning peak hour in which temporal evolution of the congestion cannot be ignored.

As shown in Proposition 3, under SO toll, the ridesharing price becomes a flat rate \$5.86 (see Fig. 6(b)). Since there is no congestion at SO, the parameter setting for ω_t and ω_d becomes more flexible. Any combination which satisfies the condition $\omega_t + 40\omega_d = 11.73$ can support the SO solution as an equilibrium.

We also examine individual travel cost in Table 5. The generalized cost is \$19.34 when there is no ridesharing, and it is \$16.99 when ridesharing is present, which verifies that commuters are better off in the presence of ridesharing. It corresponds to the comparison of Eqs. (33) and (8) in Section 3.2. The generalized travel cost is \$15.49 at SO (Eq. (53)). It indicates that the proposed combined toll and ridesharing pricing can further improve individuals' welfare in morning commute problem. It is consistent with the conclusion in Section 4.

5.3. Sensitivity analysis

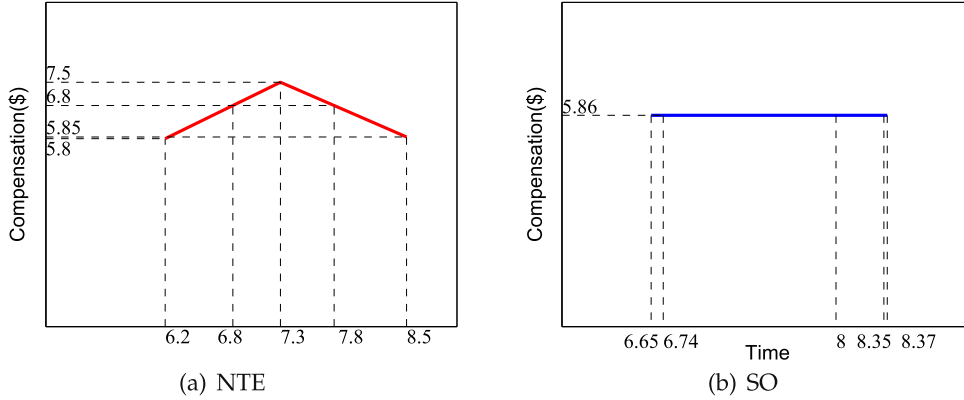
5.3.1. Sensitivity analysis at no-toll equilibrium

In this section, to test the robustness of our numerical results, we will conduct the sensitivity analysis by varying the input parameters of inconvenience cost. We will show the change in ridesharing ridership and pricing scheme with differently perceived inconvenience cost by ridesharing drivers and riders.

Table 4

System costs at three scenarios (\$ · person).

	(a) Without ridesharing	(b) With ridesharing at NTE (change from (a))	(c) With ridesharing at SO (change from (a))
Cost of queuing time	5000	2956 (−40.9%)	0 (−100%)
Cost of schedule delay	7960	4010 (−49.6%)	2917 (−63.4%)
Other (Inconvenience and out-of-pocket cost)	20,720	18,724 (−9.6%)	18,021 (−13.0%)
Total	33,680	25,689 (−23.7%)	20,938 (−37.8%)

**Fig. 6.** Ridesharing compensation.**Table 5**

Individual welfare analysis (\$).

	(a) Without ridesharing	(b) With ridesharing at NTE (change from (a))	(c) With ridesharing at SO (change from (a))
Individual commute cost	19.34	16.99 (−12.2%)	15.49 (−19.9%)

We first vary two time-based parameters in inconvenience cost functions: τ_2 for drivers and τ_3 for riders. The contour for ω_t , i.e., the price charged per time unit in ridesharing program, is plotted in Fig. 7. τ_2 and τ_3 are X axis and Y axis respectively. As stated in Proposition 1, the price scheme parameter should be set as $\omega_t = \frac{1}{2}[\tau_2 + c_t - \tau_3]$. As shown, the price charged per time unit linearly increases with the difference of inconvenience costs per time unit perceived by the driver and the rider. Furthermore, we plot the contours of solo driver flow. It reveals that, in the interior-solution region with both ridesharing and non-ridesharing commuters (i.e., $N_1 \in (0, 2000)$), with higher inconvenience cost per time unit, more commuters will stay as solo drivers. And when the inconvenience per time unit by ridesharing is low enough, all the commuters choose ridesharing $N_1 = 0$ and enjoy its cost saving in out-of-pocket cost. When the inconvenience cost per time unit by ridesharing is too high to be offset by its benefit, all the commuters choose driving alone $N_1 = 2000$.

The figure also describes the interior-solution region of $N_1 \in (0, 2000)$, where interior solution exists with both ridesharing and non-ridesharing commuters. The other area is corner solution where either $N_1 = 0$ or $N_1 = 2000$. Based on Eq. (31), the expression of N_1 is $\frac{cT_0(\alpha+c_t)(\beta+\gamma)}{\beta\gamma} \left[\frac{v_f(\delta_2+\delta_3-c_d)}{c_t-\tau_2-\tau_3} - 1 \right]$.

$$N_1 > 0 \Rightarrow \tau_2 + \tau_3 > c_t - v_f(\delta_2 + \delta_3 - c_d)$$

$$N_1 < 2000 \Rightarrow \tau_2 + \tau_3 < c_t - \frac{v_f(\delta_2 + \delta_3 - c_d)}{\frac{N}{X} + 1}$$

where $X = \frac{cT_0(\alpha+c_t)(\beta+\gamma)}{\beta\gamma}$. Thus $\tau_2 + \tau_3 \in [c_t - v_f(\delta_2 + \delta_3 - c_d), c_t - \frac{v_f(\delta_2 + \delta_3 - c_d)}{\frac{N}{X} + 1}]$. Then using the inputs in Table 3, we have the interior-solution region $\tau_2 + \tau_3 \in (1.76, 2.56)$. In order to successfully match riders with drivers (the region with positive N_2), it is necessary to constrain N_1 within $[0, 2000]$. Thus, the feasible region with a positive ridership ($N_2 > 0$) $\tau_2 + \tau_3 \in (0, 2.56)$ at NTE.

We now vary two distance-based parameters in inconvenience cost functions: δ_2 for drivers and δ_3 for riders. All the other parameters are fixed. The contours for ω_d , i.e., the price charged per distance unit in ridesharing program, are plotted in Fig. 8. δ_2 and δ_3 are X axis and Y axis respectively. Recall Proposition 1, the price charged per distance unit is determined

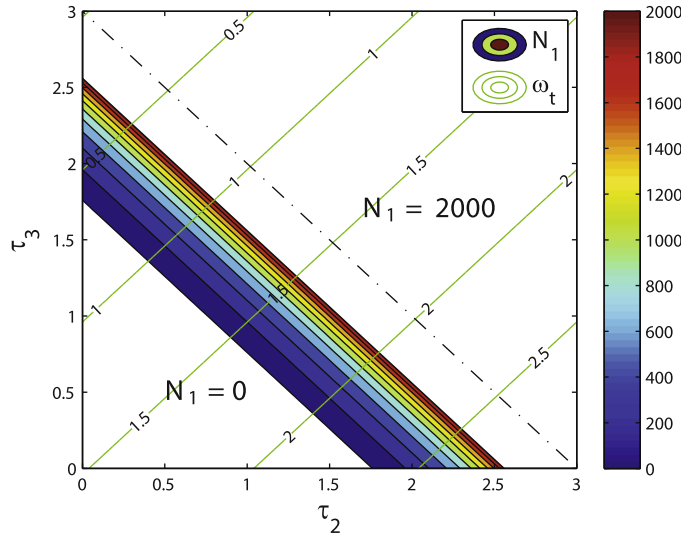


Fig. 7. Contours of ω_t and N_1 at no-toll equilibrium.

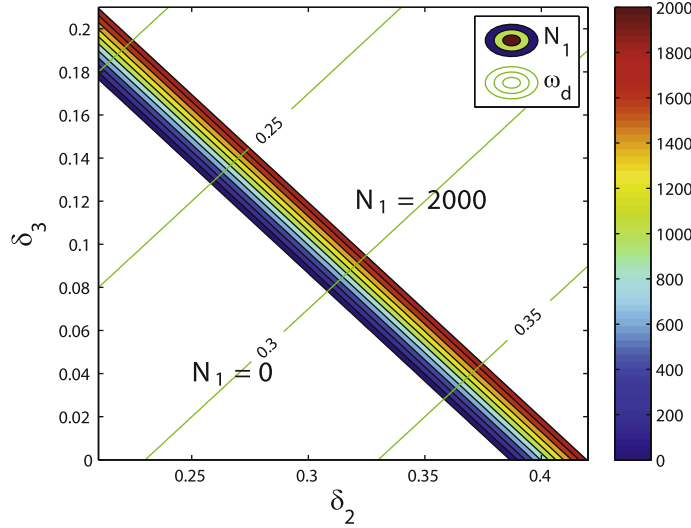


Fig. 8. ω_d and N_1 at no-toll equilibrium.

by $\omega_d = \frac{1}{2}(\delta_2 + c_d - \delta_3)$. Therefore, the price charged per distance is linearly increasing with the difference of δ_2 and δ_3 . Similarly, the corresponding interior-solution region of δ_2 and δ_3 satisfying interior solutions is $\delta_2 + \delta_3 \in (c_d + \frac{c_t - \tau_2 - \tau_3}{v_f}, c_d + (\frac{N}{X} + 1) \frac{c_t - \tau_2 - \tau_3}{v_f})$. Specifically, in this example, $\delta_2 + \delta_3 \in (0.39, 0.42)$. The contours of N_1 are plotted as well, which show the feasible region with positive ridership $N_1 \in [0, 2000]$. The feasible region with positive ridership is $\delta_2 + \delta_3 \in (0, 0.42)$. Fig. 8 also shows that with higher inconvenience cost per distance, it will result in higher solo driver flow, and thereby lower market penetration for ridesharing program.

5.3.2. Sensitivity analysis at SO

Similar to the no-toll equilibrium in the previous section, we further conduct the sensitivity analysis at SO. Fig. 9 plots the contours of solo driver flow by varying τ_2 (X-axis) and τ_3 (Y-axis). At SO, the interior-solution region becomes $\tau_2 + \tau_3 \in (c_t + (c_d - \delta_2 - \delta_3)v_f, c_t + (c_d - \delta_2 - \delta_3)v_f + \frac{N}{c f_0} \frac{\beta \gamma}{\beta + \gamma})$. With the input in Table 3, $\tau_2 + \tau_3 \in (1.76, 17.68)$. The feasible region with positive ridership is $\tau_2 + \tau_3 \in (0, 17.68)$. Compared with the feasible region at no-toll equilibrium, the upper bound of the region with positive ridesharing ridership is relaxed to 17.68 at SO. Therefore, combined with SO toll, the com-

muters are more tolerant to the inconvenience cost, and thereby it is easier to match riders with drivers even if τ_2 and τ_3 are relatively high. Solo driver flow still linearly increases with the inconvenience cost per time unit.

By varying the distance-based parameters, δ_2 and δ_3 , the contours of N_1 are plotted in Fig. 10. The feasible region for a positive ridership that satisfies interior solutions is $\delta_2 + \delta_3 \in \left(c_d + \frac{c_t - \tau_2 - \tau_3}{v_f}, c_d + \frac{c_t - \tau_2 - \tau_3}{v_f} + \frac{N}{cT_0} \frac{\beta\gamma}{\beta+\gamma} \frac{1}{v_f} \right)$. That is, (0.39, 0.78) here. The feasible region with positive N_2 is also enlarged compared with the region at no-toll equilibrium. It implies that a combined congestion pricing and ridesharing pricing will allow a larger feasible region, in terms of the perception of the convenience in distance traveled. The ridesharing operator may also benefit from the congestion pricing policy by having a more flexible compensation scheme and by attracting higher ridership for ridesharing program. It is consistent with our conclusion Proposition 3. Any combination of ω_t and ω_d that satisfies the condition can support SO, which actually results in a flat rate in morning commute. The flat rate p at SO is plotted in Fig. 11. The ridesharing compensation $p = (\omega_t + \omega_d v_f) T_0$. Let $\Delta\tau = \tau_2 - \tau_3$ and $\Delta\delta = \delta_2 - \delta_3$. Then $p = \frac{T_0}{2} [\Delta\tau + c_t + (\Delta\delta + c_d) v_f]$ according to Proposition 3. Fig. 11 demonstrates that the price p will linearly increase with perception difference in inconvenience costs by the driver and the rider.

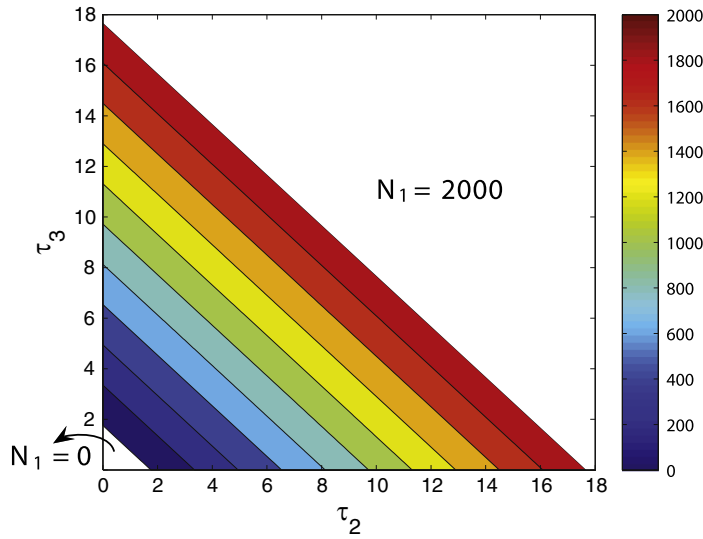


Fig. 9. Solo driver flow N_1 given τ_2 and τ_3 at SO.

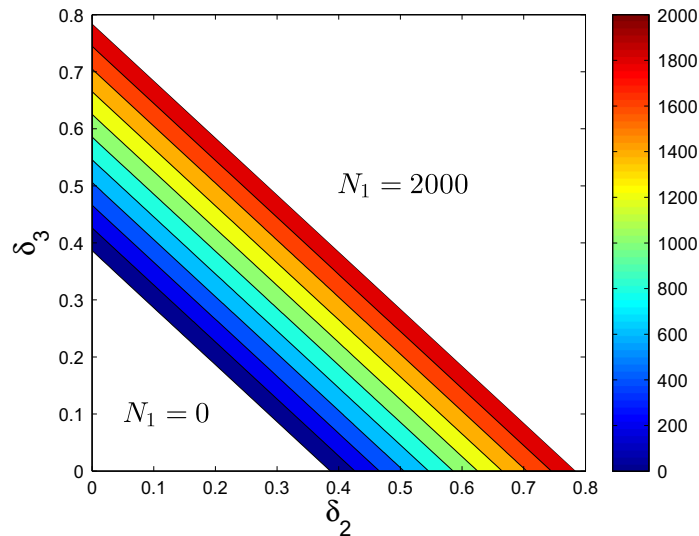


Fig. 10. Solo driver flow N_1 given δ_2 and δ_3 at SO.

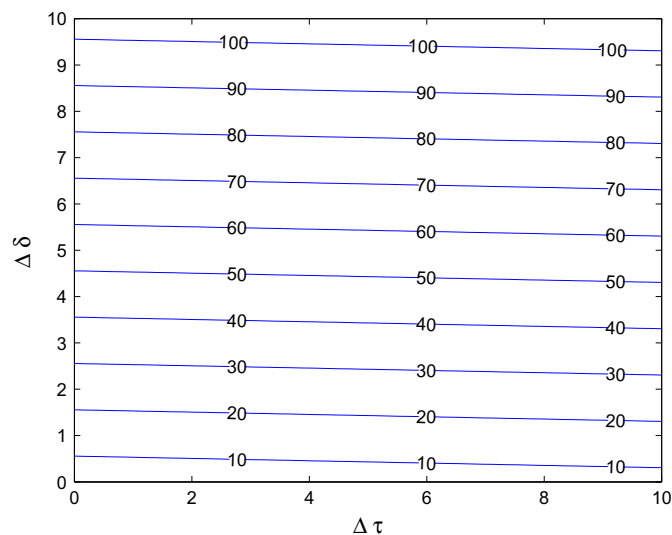


Fig. 11. The flat compensation p at SO.

6. Conclusions and future directions

This paper examines the dynamic user equilibrium of the morning commute problem in the presence of ridesharing program. Commuters simultaneously choose departure time from home and commute mode among three roles: solo driver, ridesharing driver, and ridesharing rider. We consider a combined time-based and distance-based compensation scheme, which has been practiced in ridesharing industry. A time-varying compensation scheme is proposed in order to maintain a positive ridesharing ridership at user equilibrium. We find that the compensation should be set according to the inconvenience cost functions and the out-of-pocket cost functions. Otherwise, the mismatch between the supply of ridesharing service and the demand of ridesharing service over time will occur, and nobody will choose ridesharing at the user equilibrium. The departure orders of ridesharing participants and solo drivers are discussed. When the price charged per time unit is higher than the inconvenience cost per time unit perceived by the ridesharing drivers, the ridesharing participants will travel at the center of peak hours and the solo drivers will travel at the two tails. When the price charged per time unit is lower than the inconvenience cost per time unit perceived by the ridesharing drivers, the solo drivers will travel at the center of peak hours and the ridesharing participants will travel at the two tails. Within the feasible region with positive ridership for three roles, the ridesharing program can reduce the congestion and all the commuters will be better off by introducing ridesharing program.

To achieve system optimum (SO), a time-varying toll combined with a flat ridesharing price is derived from eliminating queuing delay. The compensation becomes a flat rate at SO. Under SO toll, the ridesharing program can attract more participants and have an enlarged feasible region. That is, in some cases, the no-toll equilibrium may not have ridesharing participants, while under SO toll a positive ridership can be decentralized. This reveals that the commuters are more tolerant to the inconvenience caused by sharing a ride at SO because of the lower travel time. Compared with no-toll equilibrium, both overall congestion and individual travel cost are further reduced at SO. This study provides policy insights for the regulation of ridesharing in the morning peak hours. A combined ridesharing pricing scheme and toll scheme is proposed for optimal system performance. The numerical experiments demonstrate the ridesharing pricing schemes at no-toll equilibrium and system optimum respectively. The sensitivity analysis shows the feasible region with positive ridership for the three roles. And it also shows the changes in ridership and compensation when varying the parameters in the inconvenience cost function.

There are some limitations in the model setting. Homogeneous travel preference is considered in terms of value of time, unit cost of schedule delay, desired arrival time, and the inconvenience cost per time or distance. This assumption ignores the fact that different preferences may result in different decision-making towards ridesharing. Therefore, the possible extensions in the future study are identified. First, an elastic function for the total demand may be considered so that the reduced travel cost may attract more commuters to traverse the corridor. Second, interactions among multiple origin-destination pairs may be modeled in a network setting. Third, public transit mode, such as metro line, may be added to examine a multi-modes transportation system. Fourth, commuters with heterogeneous schedule delay cost, inconvenience cost or desired arrival time may be considered.

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